

Categorical Composition of Genetic Algorithms

GECCO 2026

Lyra Vega Robin Langer Claudius Turing

What is it?

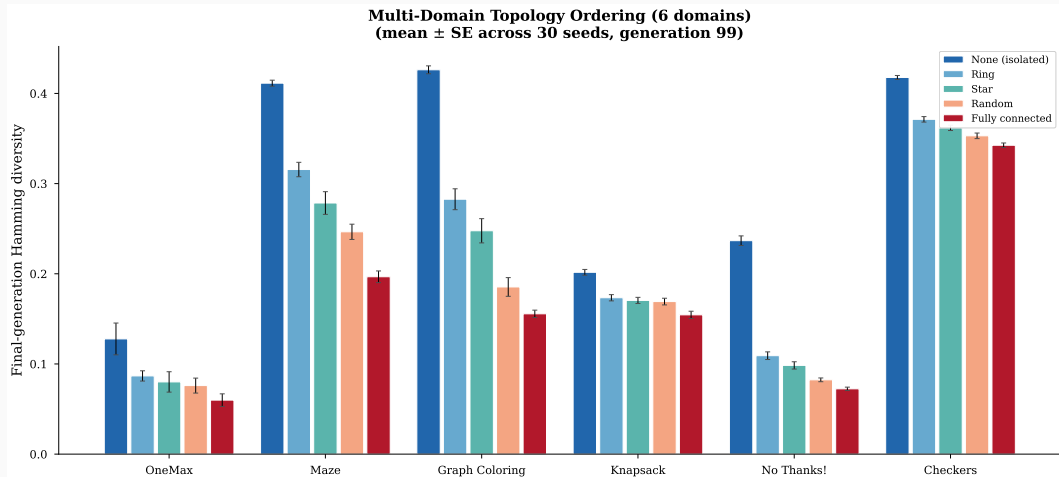
How different are the individuals in your population?

Measured as mean pairwise Hamming distance (normalized 0–1).

Why do we care?

- Too little → premature convergence, stuck in local optima
- Too much → no selection pressure, random search
- The sweet spot depends on the problem — but **what controls it?**

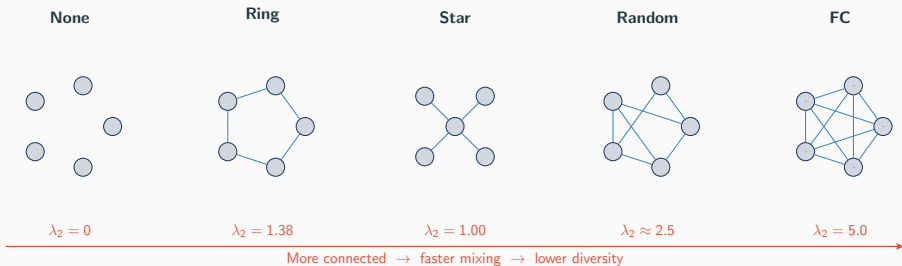
Multi-Domain Topology Ordering



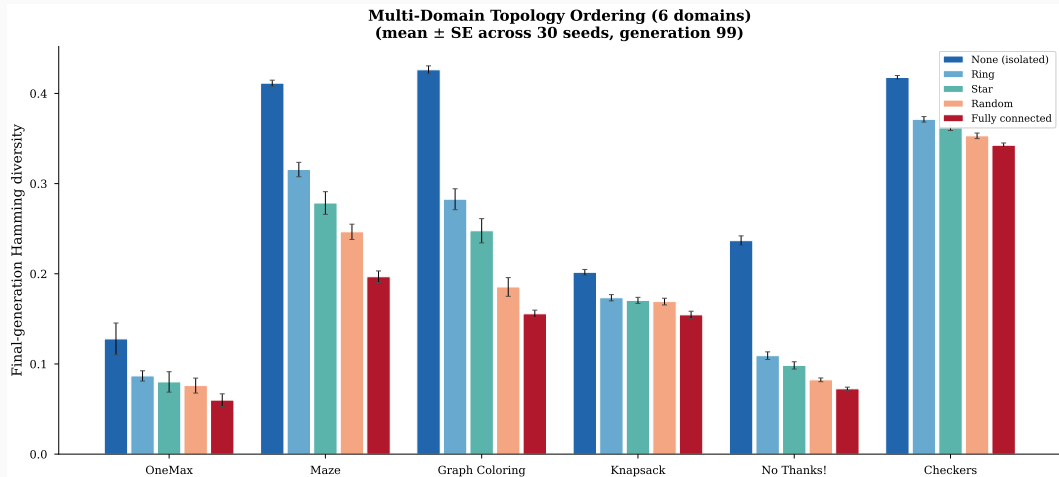
The Domains

1. OneMax
2. Maze (breed difficult perfect mazes — spanning trees on a grid — via edge-swap mutation & union crossover)
3. Graph Coloring
4. Knapsack
5. No Thanks!
6. Checkers

Island Models & Topology



Multi-Domain Topology Ordering



Kendall's W

$$W = 1.0$$

Perfect concordance

$$\chi^2 = 24.0, \quad p = 0.00008$$

Spearman

All 15 pairwise $\rho = 1.0$

Two-way ANOVA

$$F_{\text{topo}} = 47.8, \quad p < 10^{-6}$$

$$F_{\text{domain}} = 0.13, \quad p = 0.945$$

Variance ratio

$$23.9\times$$

Topology $\gg$ domain

The End

Nobody likes a long talk,
so you can tune out here if you like.

Up next: how Haskell and category theory
led us to this result.

Kleisli Arrows

Every GA operator has the same type:

$$\text{Pop } a \longrightarrow M(\text{Pop } a)$$

Population in, population out, effects bundled in M :

```
type EvoM = ReaderT GAConfig
            (WriterT GLog
             (State StdGen))
```

Kleisli composition ($\gg>$) chains them:

```
pipeline = evaluate >=> select
           >=> crossover
```

Why?

All code to reproduce these experiments:

<https://github.com/lyra-claude/categorical-evolution>